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 **Meta Review Status:** The meta review and decision have not been released yet.

 AI Review from DeepReviewer will be automatically processed

 Summary

This paper introduces the Asking Across Needs (AAN) framework and the UsClarification for Asking Needs (U-CAN) benchmark, aiming to enhance the ability of Large Language Models (LLMs) to ask effective clarification questions in human-computer interactions (HCI). The

authors argue that existing methods often focus on single needs, leading to unmet task-oriented needs and performance gaps in real-world settings. The U-CAN benchmark is designed to address these limitations by incorporating diverse user needs, interaction styles, and interaction types, and it includes both task-oriented and retrieval-related clarification scenarios. The paper evaluates several existing methods on the U-CAN benchmark and proposes a new method called User-Guided Asking Across Needs, which leverages user-guided prompting and interaction graphs to improve the performance of LLMs. The empirical findings suggest that the proposed method outperforms existing approaches, particularly in real-world scenarios with enrolled students. However, the paper's contributions are somewhat undermined by a lack of detailed explanations and thorough analysis, which I will discuss in the weaknesses section.

✓ Strengths

The paper makes several valuable contributions to the field of human-computer interaction (HCI) and the development of LLMs for clarification tasks. One of the key strengths is the introduction of the Asking Across Needs (AAN) framework, which addresses both task-oriented and retrieval-related clarification in a single, cohesive framework. This is a significant improvement over previous works that often focus on one type of clarification, thereby providing a more holistic approach to understanding and addressing user needs. The U-CAN benchmark is another notable contribution, as it provides a dataset that reflects real-world HCI scenarios, which is crucial for developing and testing more effective LLMs for clarification. The benchmark includes a diverse set of interactions, with 491 enrolled students and 352 tutors for the task-oriented need, and 561 enrolled students and 496 tutors for the retrieval-related need, each containing initial and subsequent interactions. The authors also demonstrate the effectiveness of their proposed method, User-Guided Asking Across Needs, through a series of experiments that show promising results in both simulated and real-world settings. The paper's empirical findings, particularly the performance improvements in the enrolled student setting, highlight the potential of user-guided prompting and interaction graphs in enhancing LLMs' ability to handle complex, multi-need interactions.

✗ Weaknesses

Despite the paper's valuable contributions, several weaknesses significantly impact its overall clarity and robustness. First, the paper lacks a clear and detailed explanation of the U-CAN benchmark. While the authors describe the benchmark's components, including the number of participants and the types of needs, they do not provide a comprehensive description of the data collection process, the specific criteria used to determine successful clarification, or a full list of evaluation metrics. For example, the paper mentions 'user satisfaction, clarification, and helpfulness' as evaluation metrics but does not detail how these are measured or what specific criteria are used to assess them. This lack of detail makes it difficult to assess the validity and generalizability of the benchmark, and it hinders the reader's ability to fully understand the scope and quality of the data (Section 2.3, U-CAN BENCHMARK). I am confident in this assessment based on the limited information provided in the paper. Second, the paper does not provide a clear and detailed explanation of the User-Guided Asking Across Needs method. The method is described through its components—user-guided prompting and the interaction graph designer—but the paper lacks specific examples of the prompts used, the exact format of the user guide and interaction graph data, and a detailed explanation of how these components are integrated into the LLM's decision-making process. For instance, the paper mentions prompting the LLM with `` and `` but does not specify how user background information is formatted and used to generate user-guided clarification questions (Section 2.2, ASKING ACROSS NEEDS FRAMEWORK). This lack of detail makes it challenging to reproduce the results or to understand the method's strengths and limitations. I am highly confident in this assessment, as the paper's descriptions are high-level and lack the necessary technical depth. Third, the paper lacks a thorough analysis of the results. While the authors provide some analysis and draw conclusions, they do not conduct a detailed error analysis, an ablation study to evaluate the impact of different components, or a comprehensive comparison with other state-of-the-art methods. For example, the paper does not discuss the types of errors made by the proposed method, nor does it analyze the impact of different user guidance strategies on the overall performance (Section 3, EXPERIMENTS AND ANALYSES). This omission limits the paper's ability to provide valuable insights into the method's behavior and its practical implications. I am highly confident in this assessment, as the paper's analysis is more focused on comparing different settings and models rather than a deep dive into the method's performance. Fourth, the paper does not provide sufficient evidence to support the claims made. While the authors present empirical results and comparisons with some baselines, they do not compare their proposed method with all state-of-the-art methods in the field of clarification

question generation or HCI. Additionally, the paper lacks a discussion of the computational cost associated with the proposed interaction graph designer and user-guided prompting techniques. For instance, the paper does not specify the computational resources required to train and deploy the model, nor does it discuss the feasibility of using the model in resource-constrained environments (Section 3, EXPERIMENTS AND ANALYSES). This lack of evidence and discussion makes it difficult to assess the practical value of the proposed method. I am highly confident in this assessment, as the paper's empirical results are limited in scope and do not provide a comprehensive evaluation. Fifth, the paper's focus on specific real-world scenarios, such as tutoring, may limit the generalizability of the findings. The framework's effectiveness in other domains or with different user groups is not explored, potentially narrowing its applicability. For example, the paper does not discuss how the framework performs in customer service or personal assistant contexts, nor does it address potential cultural or communication style differences (Section 2.3, U-CAN BENCHMARK). This limitation is significant, as the applicability of the framework to diverse scenarios is crucial for its broader impact. I am highly confident in this assessment, as the paper's experimental setup is primarily focused on the tutoring domain. Finally, the paper does not provide a detailed discussion of the limitations of the proposed approach. This includes a lack of discussion on potential biases in the dataset and how they might affect the performance of the model, as well as the limitations of the evaluation metrics used. For example, the paper does not analyze how the model handles ambiguous or contradictory user inputs that may not fit well into the predefined interaction graphs (Section 3, EXPERIMENTS AND ANALYSES). Addressing these limitations would make the paper more transparent and help readers understand the scope of the proposed approach. I am highly confident in this assessment, as the paper's discussion of limitations is minimal and does not cover these critical aspects.

Suggestions

To address the identified weaknesses and enhance the paper's clarity and robustness, I recommend several concrete, actionable improvements. First, the paper should provide a much more detailed description of the U-CAN benchmark. This should include a clear explanation of the data collection process, the types of tasks included (e.g., specific examples of task-oriented and retrieval-related scenarios), and the evaluation metrics used. For example, the authors should specify how the clarification questions are generated, what criteria are used to determine if a clarification is successful, and how user satisfaction is measured. Additionally, the paper should include examples of the benchmark data to illustrate the types of interactions being evaluated. This would greatly enhance the reader's understanding of the benchmark and its potential uses (Section 2.3, U-CAN BENCHMARK). Second, the paper should provide a more detailed explanation of the User-Guided Asking Across Needs method. This should include a clear description of how user guidance is incorporated into the prompting process, what types of prompts are used, and how the method handles different user needs. For instance, the authors should specify how the user's background information is used to generate user-guided clarification questions and how the interaction graph is used to guide the LLM's actions. The paper should also include a comparison of the proposed method with other prompting techniques to demonstrate its effectiveness. Without these details, it is difficult to understand the novelty and practical value of the proposed method (Section 2.2, ASKING ACROSS NEEDS FRAMEWORK). Third, the paper should provide a more thorough analysis of the results. This should include a detailed error analysis, an ablation study to evaluate the impact of different components of the proposed method, and a comparison with other state-of-the-art methods. The authors should also discuss the limitations of the proposed method and suggest directions for future research. For example, the paper should analyze the types of errors made by the proposed method, discuss the impact of different user guidance strategies on the overall performance, and explore the computational cost of the proposed method. A more detailed analysis of the results would greatly enhance the paper's credibility and provide valuable insights for future research (Section 3, EXPERIMENTS AND ANALYSES). Fourth, the paper should include a more diverse set of scenarios in the experiments to address concerns about generalizability. While the student-tutoring context is a good starting point, the paper should also explore how the framework performs in other domains, such as customer service or personal assistants. This could involve using datasets from these domains or creating new ones that reflect the specific challenges of these interactions. Additionally, the paper should discuss the potential limitations of the framework in different cultural contexts or with different user groups. For example, the way users formulate questions and the types of clarifications they need may vary across cultures. Addressing these issues would strengthen the paper's claims and make the framework more broadly applicable (Section 2.3, U-CAN BENCHMARK). Finally, the paper should provide a more detailed discussion of the limitations of the proposed approach. This should include a discussion of potential biases in the dataset and how they might affect the performance of the model, as well as the limitations of the evaluation metrics used. The authors should also discuss the potential for error propagation in the interactive graph model, since the model relies on a sequence of interactions, errors in early steps could propagate to later steps, leading to suboptimal performance. The authors should discuss how they plan to mitigate this issue and ensure the robustness of the model (Section 3, EXPERIMENTS AND ANALYSES).

? Questions

1. How well does the Asking Across Needs framework generalize to other domains beyond tutoring, such as customer service or personal assistants? Have you tested the framework in diverse settings? 2. Can you provide more details on the computational requirements of the interactive graph model? Is it feasible to deploy this model in resource-constrained environments? 3. How does the model handle ambiguous or contradictory user inputs that may not fit well into the predefined interaction graphs? 4. You mentioned using interaction graphs to model user-LLM interactions. How scalable is this approach when dealing with a large number of users or more complex interaction patterns? 5. How does the model adapt to users with different levels of expertise or different communication styles?

Scores

Soundness: **2.0**

Presentation: **1.5**

Contribution: **2.0**

Rating: 3.5

 AI Review from ZGCA will be automatically processed

Summary

The paper addresses clarification across multiple user needs in human-computer interaction (HCI), arguing that prior work mostly considers single-need scenarios. It proposes (a) the Asking Across Needs (AAN) framework that models user and LLM actions via an interaction graph inspired by Pearl (2009), (b) a new benchmark U-CAN featuring both task-oriented and retrieval-related clarification with Simulated User and Enrolled Student settings (including initial and subsequent interactions), and (c) two user-guided designs: User-Guided Askable Clarification Question (U-G-ACQ) and User-Guided Actionable Response (U-G-AR). Experiments (Tables 2–5) report that task-oriented needs are harder than retrieval-related ones, existing methods underperform with Enrolled Students compared to simulated users, and that interaction-graph-informed, user-guided designs can improve some scores in certain settings. The paper also includes real-world evaluations (Asking Clarification in the Wild; Asking Clarification with Human Annotation).

✓ Strengths

- Timely motivation: highlights a real gap between simulated and real-user settings in asking clarification and shows that task-oriented needs are often unmet (Section 3.1; Table 2).
- Benchmark contribution: U-CAN includes Enrolled Student interactions and covers both task-oriented and retrieval-related needs, with initial vs. subsequent interactions and human annotations (Section 2.3).
- Real-world evaluation: includes classroom-style evaluation and a setting that mirrors benchmark annotation (Section 3.5; Table 5).
- Unified perspective: attempts to conceptualize multi-need clarification via an interaction-graph framework (Section 2.2), encouraging a structured view of mixed-initiative HCI.
- User-guided designs (U-G-ACQ/U-G-AR) are practical directions to explore how guidance can steer models toward better clarification or actionable responses (Sections 3.3.1–3.3.2).

✗ Weaknesses

- Ambiguous core formalism: the interaction graph is defined with unclear notation (e.g., edges of the form (i, a, i) ; overloading of 'RV' for both same and different values) and no visual depiction (Section 2.2, Definitions 1–2). Δ and Δ' are introduced but not operationalized as measurable variables; this undermines the theoretical contribution and hinders reproducibility.
- Undefined metrics and procedures: key evaluation metrics (User Satisfaction, Clarification, Helpfulness) and the 'Per. "68%"/"88%" Score' columns are not defined (Tables 2–4). There are no details on rubric scales, annotation protocols, inter-annotator agreement, or quality control.
- Lack of statistical rigor: results are averages without confidence intervals, statistical significance tests, or effect sizes. This makes it difficult to assess robustness of reported gains (Tables 2–5).
- Human-in-the-loop confounders: U-G-ACQ and U-G-AR explicitly insert human-written questions/responses during the process (Sections 3.3.1–3.3.2). Without controlled comparisons, cost analyses, or ablations, it is unclear whether improvements reflect model capability or human scaffolding.
- Insufficient detail for replication: the paper lacks concrete prompts, interaction-graph instantiations, dataset release plans, and detailed descriptions of the 'Enrolled Student' protocol (participant demographics, consent, instructions, number of annotators per sample, etc.).
- Analytical depth: explanations are largely qualitative; there are no fine-grained error analyses or ablations (e.g., impact of user background vs. interaction graph; initial vs. subsequent turns; task-oriented vs. retrieval-related) beyond averages.
- Claim-evidence alignment: while Section 3.2 states prompting alone cannot meet multiple needs, the evidence is limited and lacks the methodological detail necessary to draw firm conclusions from Tables 2–3.
- Positioning vs. related work: although prior studies are cited, the paper does not clearly delineate how AAN differs formally and operationally from existing mixed-initiative and retrieval-clarification frameworks, nor does it provide canonical examples clarifying the unique contributions.

? Questions

- Formalization: Please provide a precise, unambiguous definition of the interaction graph. What is the exact meaning of edges like (i, a, i) ? How are 'UV' and 'RV' defined (and why is 'RV' used for both same and different values in Section 2.2)? Can you include a diagram of a

concrete small interaction graph tracing a full dialogue?

- Operationalizing needs: You introduce Δ and Δ' to denote task-oriented and retrieval-related need satisfaction in Section 2.2. How are these instantiated as measurable quantities in experiments? What thresholds or scoring rules map dialogue states to Δ/Δ' values?
- Metrics: How are User Satisfaction, Clarification, Helpfulness defined? Are they Likert scales with explicit rubrics? Who annotates them (end users vs. third-party annotators), and what training/guidelines did annotators receive?
- Reliability: What are the inter-annotator agreement statistics (e.g., Cohen's κ or Krippendorff's α) for each metric? How many annotators per instance? Were disagreements adjudicated?
- Statistics: Please report confidence intervals and statistical significance testing for all key comparisons (e.g., GPT-4 vs. GPT-4 + U-G-ACQ/U-G-AR across Table 4 and Table 5).
- 'Per. "68%"/"88%" Score': What do these scores represent in Table 4? How are they computed and why those thresholds?
- Human-guided designs: For U-G-ACQ and U-G-AR (Sections 3.3.1–3.3.2), please detail the exact role of human-written guidance: at which turns is it injected, by whom, and using which protocol? Provide an ablation isolating the impact of the human guidance vs. model prompting alone, including time/cost.
- Prompts and reproducibility: Can you share the full prompts for user-guided background and interaction-graph conditioning? What temperature/top-k settings and model versions were used?
- Dataset details: For U-CAN, please provide dataset size per split, domain coverage, demographic info for Enrolled Students (if any), privacy safeguards, consent procedures/IRB, and plans for release (data/licensing, annotation guidelines, evaluation scripts).
- Baselines: Beyond GPT-4 and a few listed systems, can you include stronger retrieval-clarification baselines (e.g., CLARINET) and more recent multi-turn agents? Also clarify how code models (e.g., CodeGen) were adapted for task-oriented dialogues.
- Real-world settings: For Section 3.5, please detail the classroom setup, number of participants, the exact evaluation interface, whether students saw multiple system outputs, and how order/position biases were controlled.
- Case studies: Could you include a few fully worked-out dialogue traces illustrating how the interaction graph guided decisions and where U-G-ACQ or U-G-AR concretely improved outcomes?

Limitations

- Framework ambiguity: The current AAN interaction-graph formalism is difficult to interpret and reproduce without clearer notation and visuals (Section 2.2).
- Evaluation incompleteness: Metrics are undefined; there are no IAA statistics or significance tests; 'Per. "68%"/"88%" Score' is unexplained (Tables 2–4).
- Human-in-the-loop confounding: U-G-ACQ/U-G-AR rely on human-written questions/responses, making it hard to attribute gains to the proposed modeling vs. human assistance and raising cost/scalability concerns.
- Generalization: The paper reports averages without ablations or domain-specific breakdowns, limiting insight into which sub-cases benefit and how findings transfer.
- Resource transparency: Missing details on prompts, hyperparameters, model versions, and compute budgets limit reproducibility.
- Potential societal impacts: Deploying systems that ask proactive clarifications can inadvertently elicit sensitive information or reflect annotator/user-group biases; privacy and consent procedures for Enrolled Students should be clearly documented.

Image Evaluation

Cross-Modal Consistency: 28/50

Textual Logical Soundness: 16/30

Visual Aesthetics & Clarity: 10/20

Overall Score: 54/100

Detailed Evaluation (≤500 words):

Visual ground truth (tables only)

- Table 1: Benchmark comparison across user grouping, styles, types; cells coded "55/51"; U-CAN marked "Multiple".
- Table 2: Preliminary results (US, C, H) across models; GPT-4 tops, CodeGen lowest.
- Table 3: User-Guided Prompt results (human-only) for GPT-4, M2Lingual; small deltas.
- Table 4: UG-ACQ/UG-AR vs GPT-4 across needs; sections for Human and Automated; unclear "Per. '68%/88%'" columns; split by Initial/Subsequent.
- Table 5: Real-world (ASHT, SH) human scores; U-G-ACQ stable; U-G-AR worse.

1. Cross-Modal Consistency

- Major 1: Text repeatedly cites a missing "Example" visual for core concepts and workflows, blocking verification. Evidence: "Example shows a human-annotated example..."; "Based on the interaction graph in Example...".
- Major 2: Table misreference in Sec. 3.2 (points to Table 2 but describes user-guided prompt results shown in Table 3). Evidence: "The results are shown in Table 2."
- Major 3: Metric naming inconsistent across tables, hindering comparisons (US/C/H vs "Tutor Satisfaction", "Per. '68%' Score"). Evidence: Table 2 headers "US/C/H"; Table 4 headers include "Tutor Satisfaction" and "Per. '68%' Score".
- Minor 1: Acronym drift (ASHT/"in the Wild", SH) and model name inconsistencies (M2lingual vs M2Lingual).
- Minor 2: Table 1 coding "55/51" is nonstandard and loosely mapped to satisfied/unsatisfied.

2. Text Logic

- Major 1: Formalism inconsistency for edges and actions (edges defined among variables, later treated as triplets with actions). Evidence: "E represents the set of directed edges among these variables." vs "adding a new edge ($i^{(m-1)}$, $a^{(m)}$, $i^{(m)}$)".
- Major 2: Claims about human vs automated gap in Sec. 3.5 cannot be supported by Table 5 (human-only). Evidence: Table 5 caption: "The scores are all from human annotation."
- Minor 1: Contradictory phrasing on scores being "much higher compared to... GPT-4" within the same section; weakly supported by small 0.2 deltas.
- Minor 2: Multiple typos and duplicated terms ("timSe", "RepphraseAndRespond", "unmet unmet") disrupt clarity.

3. Figure Quality

- Major 1: Ambiguous columns ("Per. '68%/88%' Score" in Table 4) lack definitions, obscuring interpretation. Evidence: Table 4 headers include "Per. '68%' Score" and "Per. '88%' Score".
- Major 2: Table 1's binary code (55/51) is opaque; legend insufficient for quick reading. Evidence: "Table 1: ... '55' means 'not satisfied' and '51' means 'satisfied'."
- Minor 1: Sectioned rows inside Table 4 ("Automated Scores (Initial/Subsequent)") crowd readability; clearer separation or separate tables recommended.
- Minor 2: Column labels switch between "Clarification", "Helpfulness", and "Tutor Satisfaction" without explicit mapping.

Key strengths:

- Timely problem framing: multi-need HCI with interaction graphs.
- Inclusion of human-in-the-loop settings (Enrolled Students) and real-world evaluations.
- Attempt to disentangle ACQ vs actionable responses.

Key weaknesses:

- Missing core figures; formalism inconsistency blocks method comprehension.
- Table references and metric names are inconsistent, hindering claim verification.
- Ambiguous/undefined columns and unconventional coding reduce readability.
- Several claims (human vs automated gaps) lack proper supporting evidence.



Scores

Originality: 2

Quality: 2

Clarity: 2

Significance: 2

Soundness: 2

Presentation: 2

Contribution: 2

Rating: 4

 Summary

This paper introduces the Asking Across Needs (AAN) framework and the UsClarification for Asking Needs (U-CAN) benchmark, aiming to address the complexities of human-computer interaction (HCI) in the context of clarification question generation. The core idea revolves around using interaction graphs, inspired by Pearl's causality framework, to model the interplay between user actions and large language model (LLM) responses. The authors propose user-guided prompting techniques, including User-Guided Askable Clarification Question (U-G-ACQ) and User-Guided Actionable Response (U-G-AR), to enhance the quality of clarification questions. The U-CAN benchmark is presented as a novel dataset designed to capture real-world HCI scenarios, featuring both task-oriented and retrieval-related clarification needs. The experimental evaluation involves a preliminary investigation of existing methods, followed by an assessment of the proposed user-guided prompting techniques. The results, measured using human and automated scores for User Satisfaction (US), Tutor Satisfaction (TS), and Helpfulness (H), suggest that the proposed methods can improve the quality of clarification questions, particularly in the context of task-oriented needs. However, the paper also acknowledges that the performance of the proposed methods is not consistently superior to that of commercial GPT-4 across all metrics and settings. The authors emphasize the importance of considering multiple user needs and the dynamic nature of HCI, which they argue is often overlooked in existing benchmarks. The paper's contribution lies in its attempt to bridge the gap between simulated and real-world HCI scenarios through the AAN framework and the U-CAN benchmark. However, the paper also faces several challenges, including the limited scope of the experimental evaluation, the lack of detailed descriptions of the benchmark, and the need for more robust justification of the proposed methods. The paper's overall significance is in its attempt to address the complexities of HCI in the context of clarification question generation, but it also highlights the need for further research in this area.

 Strengths

The paper's primary strength lies in its attempt to address a critical gap in existing research on clarification question generation: the need to consider multiple user needs and the dynamic nature of human-computer interaction. The introduction of the Asking Across Needs (AAN) framework and the UsClarification for Asking Needs (U-CAN) benchmark represents a novel approach to modeling HCI using interaction graphs. This framework, inspired by Pearl's causality framework, provides a structured way to represent the interplay between user actions and LLM responses. The authors' focus on real-world HCI scenarios, as opposed to purely simulated environments, is another significant strength. The U-CAN benchmark, with its inclusion of both task-oriented and retrieval-related clarification needs, aims to provide a more comprehensive evaluation of clarification methods. The paper also introduces user-guided prompting techniques, such as User-Guided Askable Clarification Question (U-G-ACQ) and User-Guided Actionable Response (U-G-AR), which demonstrate the potential for improving the quality of clarification questions. The experimental results, while not consistently superior to commercial GPT-4, do show that these techniques can enhance user satisfaction in certain contexts. The paper's emphasis on the limitations of existing benchmarks, which often focus on single-turn interactions and simulated users, is also a valuable contribution. By highlighting these limitations, the authors underscore the need for more realistic and comprehensive evaluation methods. The paper's attempt to bridge the gap between simulated and real-world HCI scenarios is a significant step forward in this field. The authors also provide a preliminary investigation of existing methods, which helps to contextualize the performance of the proposed techniques. Overall, the paper's strengths lie in its novel framework, its focus on real-world scenarios, and its attempt to address the complexities of HCI in the context of clarification question generation.

 Weaknesses

After a thorough review of the paper and the provided analyses, several significant weaknesses have been identified. Firstly, the paper suffers from a lack of clarity and organization, making it difficult to follow the authors' line of reasoning. Multiple reviewers have pointed out that the writing is unclear, and the arguments are not well-structured. This lack of clarity extends to the definitions and explanations of key concepts, such as the interaction graph and the 'need' variable. For instance, the paper does not provide a clear explanation of how the 'need' variable is quantified or how it changes over the course of an interaction. The paper also lacks a clear explanation of the relationship between task-oriented and retrieval-related needs, and why these are considered separate categories. This lack of clarity makes it difficult to understand the core contributions of the paper. Secondly, the paper's claims of novelty are questionable. While the authors introduce the AAN framework and the U-CAN benchmark, the core ideas of using interaction graphs and user-guided prompting are not entirely new. The paper does not adequately acknowledge or discuss prior work that has explored similar concepts. For example, the paper does not discuss the use of interaction graphs in other contexts or the various prompting techniques that have been proposed for generating clarification questions. This lack of engagement with existing literature weakens the paper's claims of innovation. Thirdly, the experimental evaluation is limited in scope and lacks sufficient detail. The paper primarily focuses on a preliminary investigation and the evaluation of user-guided prompting techniques, with limited comparison to a wide range of existing methods. The paper does not include ablation studies to assess the contribution of different components of the proposed methods. The evaluation is also limited to a specific dataset and does not explore the generalizability of the proposed methods to other domains or tasks. Furthermore, the paper lacks detailed descriptions of the U-CAN benchmark, including the data collection process, annotation details, and statistical analysis. This lack of transparency makes it difficult to assess the quality and reliability of the benchmark. The paper also relies heavily on subjective evaluation metrics, such as user satisfaction, without providing sufficient justification for their use. The paper does not include objective metrics that directly measure the quality of the generated clarification questions. Finally, the paper's discussion of the limitations of the proposed methods is inadequate. The authors do not fully address the potential challenges of applying their methods to different types of user needs or interaction styles. The paper also does not discuss the potential for bias in the user-guided prompting techniques. Overall, these weaknesses significantly undermine the paper's contributions and limit its impact. The lack of clarity, the questionable novelty, the limited experimental evaluation, and the inadequate discussion of limitations all contribute to a paper that falls short of its potential.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. Firstly, the authors should significantly improve the clarity and organization of the paper. This includes providing clear definitions of key concepts, such as the interaction graph and the 'need' variable, and explaining the relationship between task-oriented and retrieval-related needs. The authors should also restructure the paper to improve the flow of arguments and ensure that each section builds upon the previous one. Secondly, the authors should provide a more thorough discussion of related work, acknowledging and addressing prior research that has explored similar concepts. This includes discussing the use of interaction graphs in other contexts and the various prompting techniques that have been proposed for generating clarification questions. The authors should also clearly articulate the specific contributions of their work and how it differs from existing approaches. Thirdly, the authors should expand the scope of their experimental evaluation. This includes comparing their methods to a wider range of existing methods, conducting ablation studies to assess the contribution of different components of their methods, and evaluating their methods on multiple datasets to assess their generalizability. The authors should also include objective metrics that directly measure the quality of the generated clarification questions, in addition to subjective metrics like user satisfaction. Fourthly, the authors should provide a detailed description of the U-CAN benchmark, including the data collection process, annotation details, and statistical analysis. This will improve the transparency and reliability of the benchmark and allow other researchers to use it effectively. The authors should also consider releasing the benchmark data to the research community. Finally, the authors should provide a more thorough discussion of the limitations of their proposed methods. This includes addressing the potential challenges of applying their methods to different types of user needs or interaction styles and discussing the potential for bias in the user-guided prompting techniques. The authors should also discuss the computational cost of their methods and their scalability to larger datasets. By addressing these weaknesses, the authors can significantly improve the quality and impact of their work.

 Questions

Several key questions arise from my analysis of this paper. Firstly, how exactly is the 'need' variable quantified and operationalized within the interaction graph? The paper does not provide a clear explanation of how this variable is measured or how it changes over the course of an interaction. What are the specific criteria used to determine the value of the 'need' variable, and how are these criteria applied in practice? Secondly, what is the precise relationship between task-oriented and retrieval-related needs? The paper presents these as separate categories, but it is not clear why they are considered distinct and how they interact with each other. Is there a hierarchical relationship between these needs, or are they independent? Thirdly, what are the specific details of the user-guided prompting techniques, such as U-G-ACQ and U-G-AR? The paper provides a high-level description of these techniques, but it lacks details on how they are implemented and how they differ from existing prompting methods. What are the specific prompts used, and how were these prompts designed? Fourthly, what are the specific details of the U-CAN benchmark, including the data collection process, annotation details, and statistical analysis? The paper lacks transparency regarding the creation of this benchmark, making it difficult to assess its quality and reliability. How was the data collected, what are the characteristics of the data, and what are the statistics of the dataset? Finally, what are the limitations of the proposed methods, and how might these limitations affect their applicability to different types of user needs or interaction styles? The paper does not fully address the potential challenges of applying their methods to different contexts. What are the potential biases in the user-guided prompting techniques, and how might these biases affect the quality of the generated clarification questions? These questions highlight key uncertainties and areas where further clarification is needed to fully understand the paper's contributions and limitations.

 Scores

Soundness: 2.0

Presentation: 1.25

Contribution: 1.5

Confidence: 3.75

Rating: 2.5

Keywords

Extract Keywords

Click the button to extract keywords

Insights

Extract Insights

Click the button to extract insights